

Autodesk® Scaleform®

Font and Text Configuration Overview

Scaleform 4.2
()

Scaleform C++ API

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: 2.2
: 2012 6 21

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Autodesk® Scaleform® 4.2

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1

: Scaleform

HTML

FreeType 2

/

/

Flash® Studio
SWF

Flash

Flash

Scaleform
GFX::Translator

Gfx::FontLib

FreeType 2

Flash Studio Scaleform
Flash

Scaleform

Flash

Scaleform

UI

Flash
Flash

Flash

(Text Field)

UI

Flash

UI

1.1 Flash のテキストとフォントの基礎

Flash Studio

UI

Flash Studio

Flash

GF 1 -

1.1.1

Flash

Flash Studio []



ActionScript

GfX::Translator

2

()

a)

b)

Arial

[] []

SWF

Flash 8

()

()

()

1

SWF

Flash Player

Scaleform

()

()

[...]

1.1.2

1 TextField

textfield

(Scaleform

GFx::FontLib

[]

[...]

1		1 KB	450 K
114	– +	12 KB	480 K
596	–	70 KB	630 K
7,583	–	2,089 KB	3,500 K
18,437	–	5,131 KB	8,000 K

“Arial Unicode MS”

500

150K

1.1.4

1. UI

Flash

(

2. Gfx::FontLib

(

)
SWF/GFX

Gfx::FontLib

3.

IME

4. GfxEport

-fc
10 30%

5.3

Gfx::FontLib

IME

1.2 ゲーム UI のフォントに関する決定

Scaleform

UI

4

1:

2:

3:

Scaleform

4:

Scaleform m

UI

UI

2

3

読み込みフォントの置き換え

Gfx::FontLib

デバイスフォントエミュレーション

(

)

カスタムアセットの生成 Flash

Player

SWF/GFX

Scaleform m

3:

Scaleform m

Gfx::FontLib

FreeType 2

Gfx::FontLib

4

Scaleform

3

gfixexport

2 1:

UI

UI

UI

UI

Scaleform

fonts_en.swf fonts_kr.swf

SWF

Flash

Flash

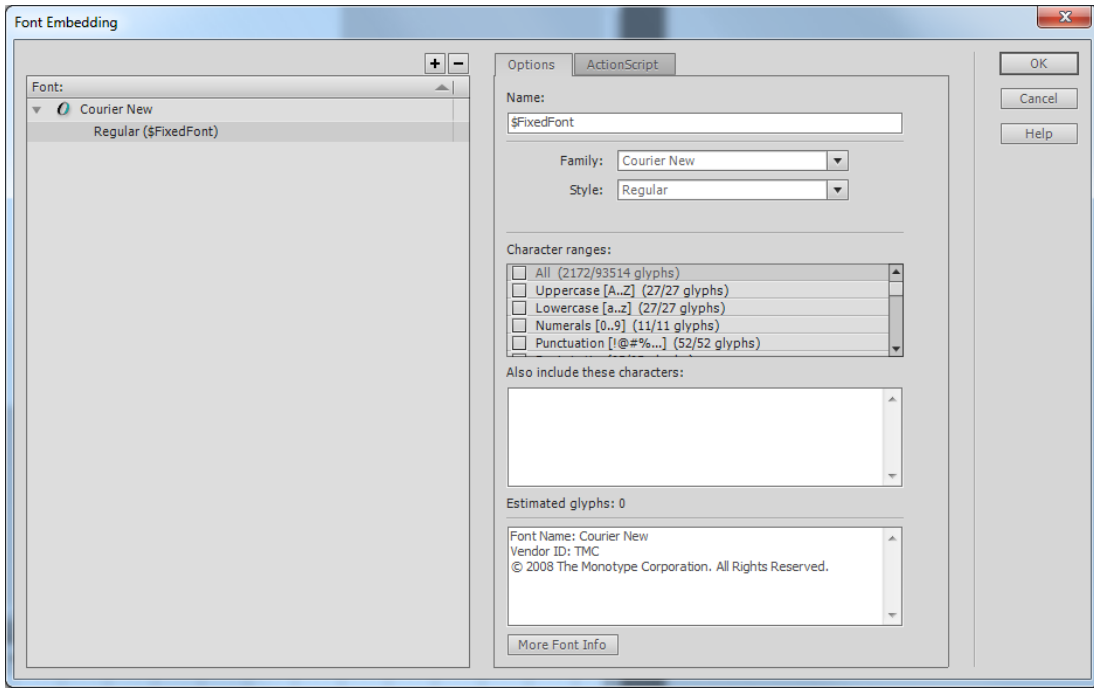
2.1 フォントシンボル

Flash Studio

W

L

[...]



FLA

“\$FixedFont”

“\$ArialGame*”

UI

‘\$’

Web

]

Flash Studio

\$

2.1.1

FLA

/

FLA

FLA

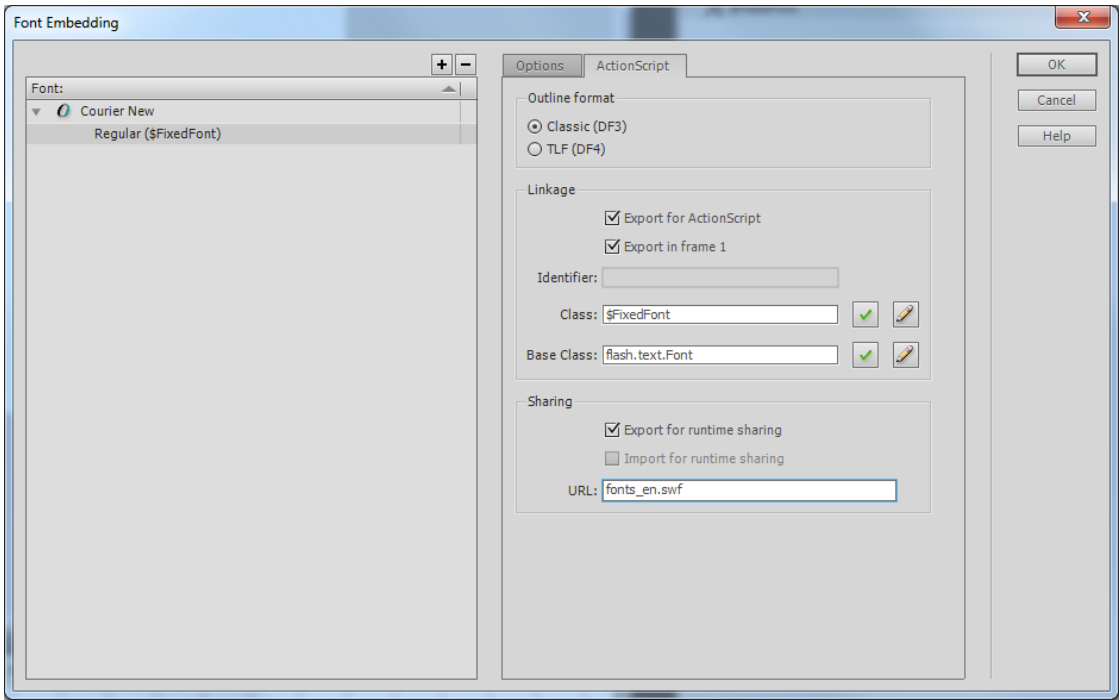
[]

[]

FLA

Flash / "Properties"

"ActionScript"



"OK" Flash

ActionScript

URL

SWF

GfX::FontLib

URL

"fonts_en.swf"

GfX

Loader.SetDefaultFontLibName("fonts_en.swf");

GfXPlayer

fonconfig.txt

fontlib "fonts_en.swf"

fontlib

" : \$ "

/

Scaleform

SWF

SWF

```

        TextField.htmlText
        ("Arial"
        )
        ActionScript
    (
    )
    TextField
    TextFormat

```

2.1.2

Flash Studio CS4

```

    Windows
    [Unicode
    243
    ]
    [
    "English"
    11,920
    "gfxfontlib.swf"
    ] > [
    ] >

```

Flash CS5
 "gfxfontlib.swf"
 fonts_en.swf fonts_jp.swf

Flash Studio CS4
 "gfxfontlib.swf"

GfX 4.0

2.2 gfxfontlib.swf を作成する手順

Flash Studio FLA

1.

'\$NormalFont'

'\$TitleFont'

2.

FLA

3.

1

4.

[]

5.

6.

fonts_<lang>.swf

lang

'en'

'ru'

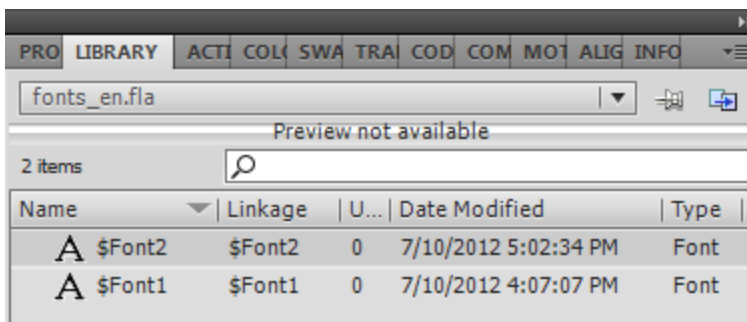
'jp'

'cn'

'kr'

FontMap

FLA



/

3 2:

UI

7. 読み込みフォントの置き換え
8. デバイスフォントエミュレーション
9. カスタムアセットの生成

Gfx::FontLib

Gfx::FontMap

'fonts_en.swf'

Gfx::Translator

Gfx::Translator

Flash Studio

UI

UI

3.1 読み込みフォントの置き換え

Flash

1

/
fonts_en.swf

UI

UI

UI

Adobe Flash Player

"fonts_en.swf"

Scaleform

```
Loader::SetDefaultFontLibName (const char* filename)          C++
                        'filename'
```

```
Loader.SetDefaultFontLibName("fonts_kr.swf");
```

```
SWF                CreateInstance
```

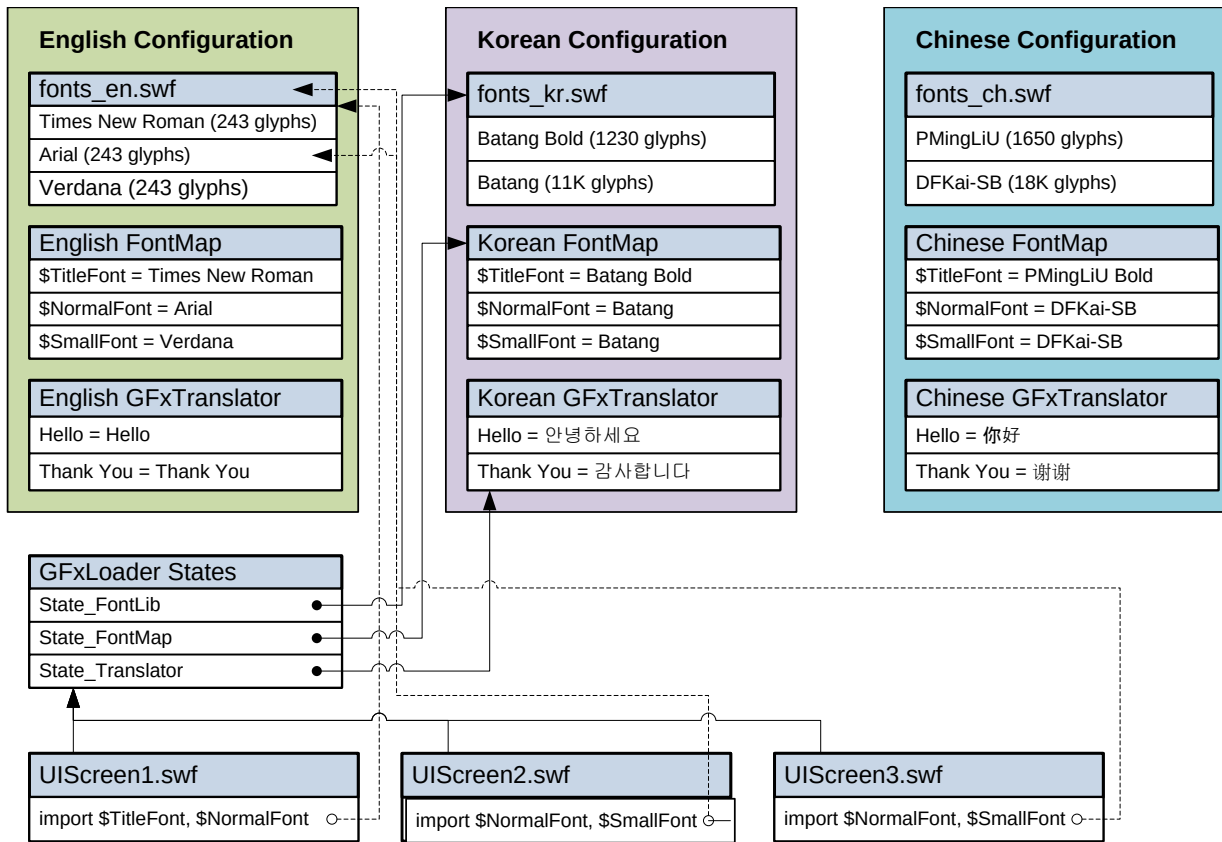
1. 文字列変換テーブル: ゲームの各フレーズを翻訳されたフレーズにマップするものです。

2. :

3. : ("NormalFont")

```
Scaleform                                Gfx::Loader
    Gfx::Translator  Gfx::FontLib        Gfx::FontMap
Gfx::Loader::CreateMovie
```

```
CreateMovie
```



“UIScreen1.swf” “UIScreen3.swf” “fonts_en.swf”

Scaleform
 GFx::FontLib 2 GFx::FontMap 1 GFx::Loader 2
 Loader::SetDefaultFontLibName()
 GFx::FontLib UI

“fonts_kr.swf”

UI

3.1.1

		SWF	
	GFX		
GfX::FontMap		GfX::Translator	
Scaleform		"fontconfig.txt"	
1.	1		"fonts_en.swf"
2.			
	GfX::Translator		
3.			GfX::FontMap
4.		"fonts_kr.swf"	
		UI	
1.		GfX::Loader	
	GfX::Translator	Translate	
2.	GfX::FontMap	GfX::Loader	MapFont
		GfX::FontMap	
	GfX::FontMap	3	
3.		(	
	GfX::FontCacheMamanger, GfX::FontProvider	GfX::FontPackParams	)
4.	GfX::FontLib	GfX::Loader	
5.		SetDefaultFontLibName(filename)	
	swf		
	Loader.SetDefaultFontLibName("fonts_en.swf");		
	GfX::FontLib		SWF/GFX
		GfX::FontLib::AddFromFile	
6.	GfX::Loader::CreateMovie		

The screenshot shows a video player window. At the top, the title bar reads "Scaleform Player (SWF/GFX Scaleform Player)". Below the title bar, the file name "fontconfig.txt" is displayed. The main area shows a video frame with a black background. Below the video frame, there is a playback control bar with a progress slider, a volume icon, and a full-screen button. At the bottom, there is a status bar with the text "F2 Ctrl+N Scaleform Player HUD" and a language selection dropdown menu showing "8 ASCII UTF-16".

```
[FontConfig "Chinese"]
fontlib "fonts_ch.swf"

map "$TitleFont" = "PMingLiU" Bold
map "$NormalFont" = "DFKai-SB" Normal
map "$SmallFont" = "DFKai-SB" Normal

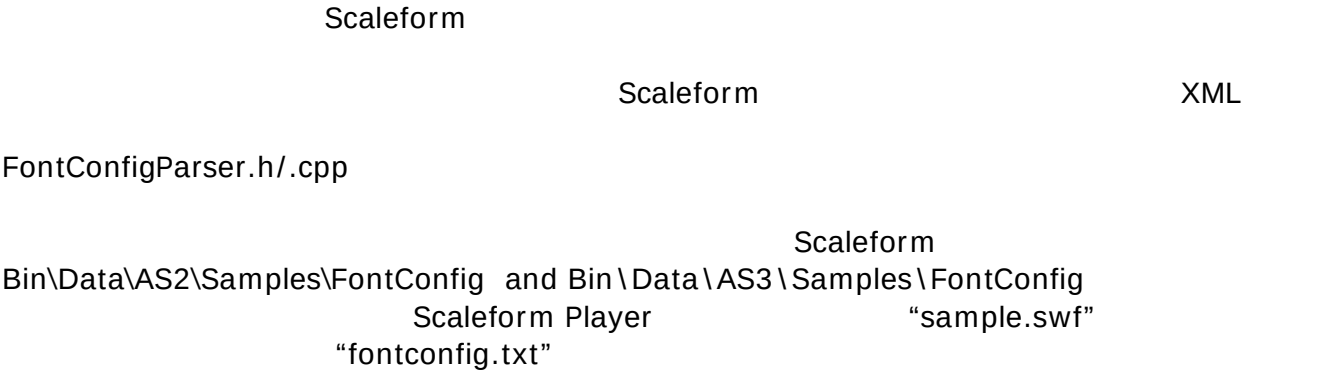
tr "Hello" = "你好"

tr "Thank You" = "谢谢"
```

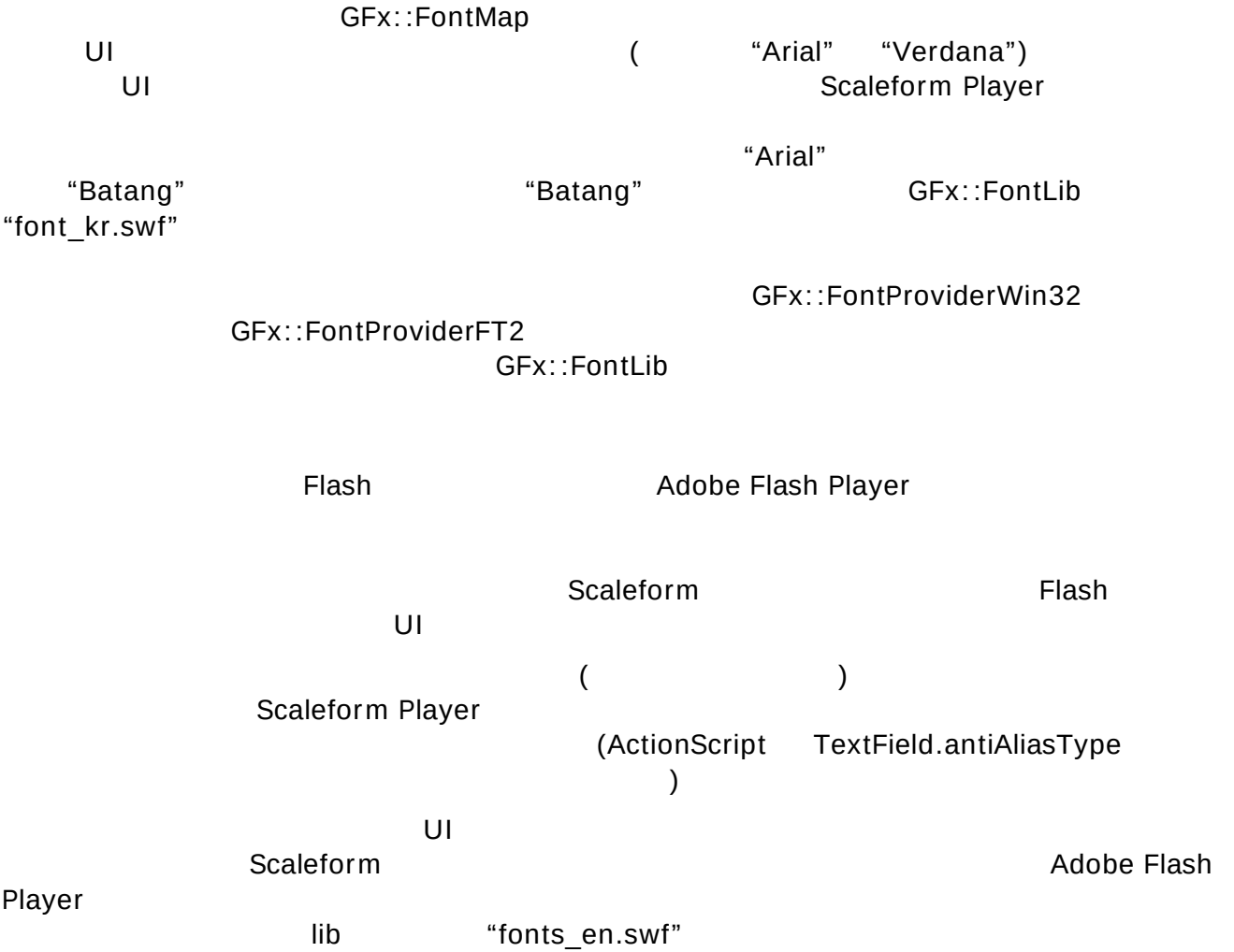
[FontConfig
"name"] Scaleform Player HUD

fontlib "fontfile"	SWF/GFX fontlib Gfx::FontLib
map "\$UIFont" = "PMingLiU"	Gfx::FontMap UI Gfx::FontMap \$UIFont "fonts_en.swf" UI

tr "Hello" = "你好"	
-------------------	--



3.1.3



8. 'Library' 'New Font...'

\$Font1	"Family"	" " "Style"
"Regular"		
9. "Character ranges" 'Korean Hangul
(All)'
10. 'ActionScript' 'Export for ActionScript' 'Export for frame 1' 'Export
for run-time sharing' 'URL' 'fonts_kr.swf'
OK
ActionScript 3.0 Flash Studio 'A definition
for this class could not be found...' 'OK'
11. 8 10 '\$Font2' 'Family'
, ,
12. 'fonts_en.swf' 'fonts_kr.swf'
13. main-app.fla.
'fonts_en.swf' '\$Font1' '\$Font2'
Ctrl-C 'Copy'
14. 'main-app.fla' j Ctrl-V
'Paste' 'Import:'
15. 'main-app.fla' 'Classic Text' 'Dynamic Text'
'Family' '\$Font1 *'
'\$TEXT1' ID
16. 2 '\$Font2 *' '\$TEXT2'
17. 'main-app.swf'
18. swf fontconfig.txt Unicode
UTF-8

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$Font1" = "Arial"
map "$Font2" = "Times New Roman"
tr "$TEXT1" = "This is"
tr "$TEXT2" = "ENGLISH!"
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$Font1" = "Batang"
map "$Font2" = "BatangChe"
tr "$TEXT1" = "이것"
tr "$TEXT2" = "은 한국이다!"
```

重要 :

“ ” “ ”
 “Batang” “BatangChe”
 SWF

Flash SWF
 fontconfig.txt

gfxexport -fntlst <swfname>

*.lst

19. Unicode UTF-8

GfXPlayer ‘main-app.swf’
 ‘English!’ Ctrl-N

fonts_kr.swf

‘This is’
 fonts_kr.swf

3.2 カスタムアセットの生成

SWF/GFX

“UIScreen.fla”
 SWF

“UIScreen.swf”

“UIScreen.swf”

(UI 600K

)

UI

UI

(
)

UI

IME

Flash Studio

[

]

(Ctrl + F11)

Scaleform

Flash

[

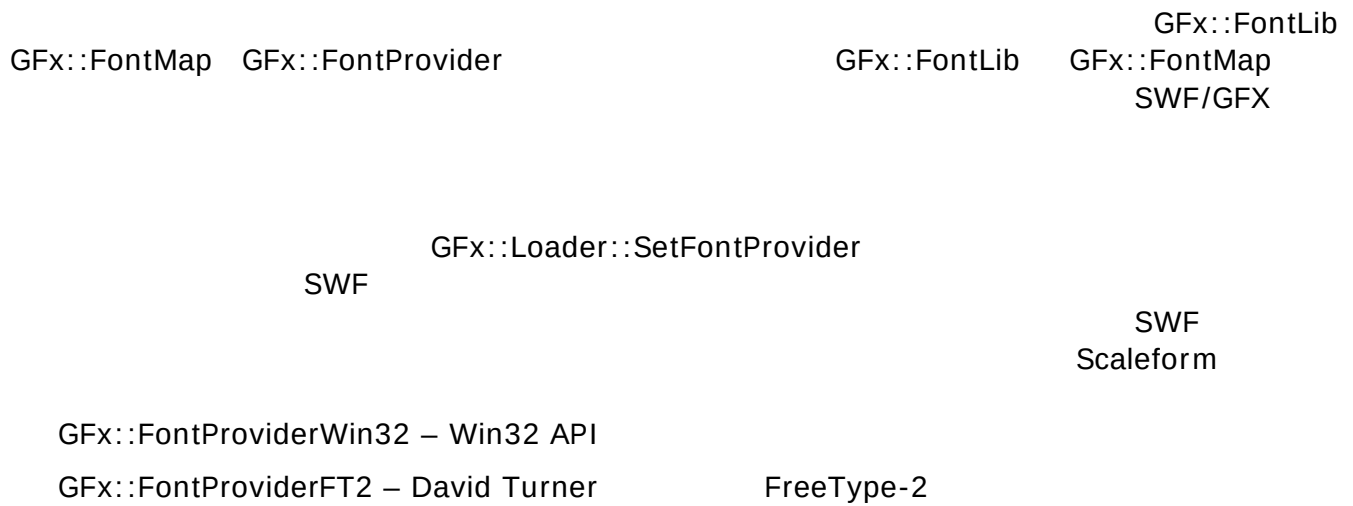
]

4 3:

Flash
HTML

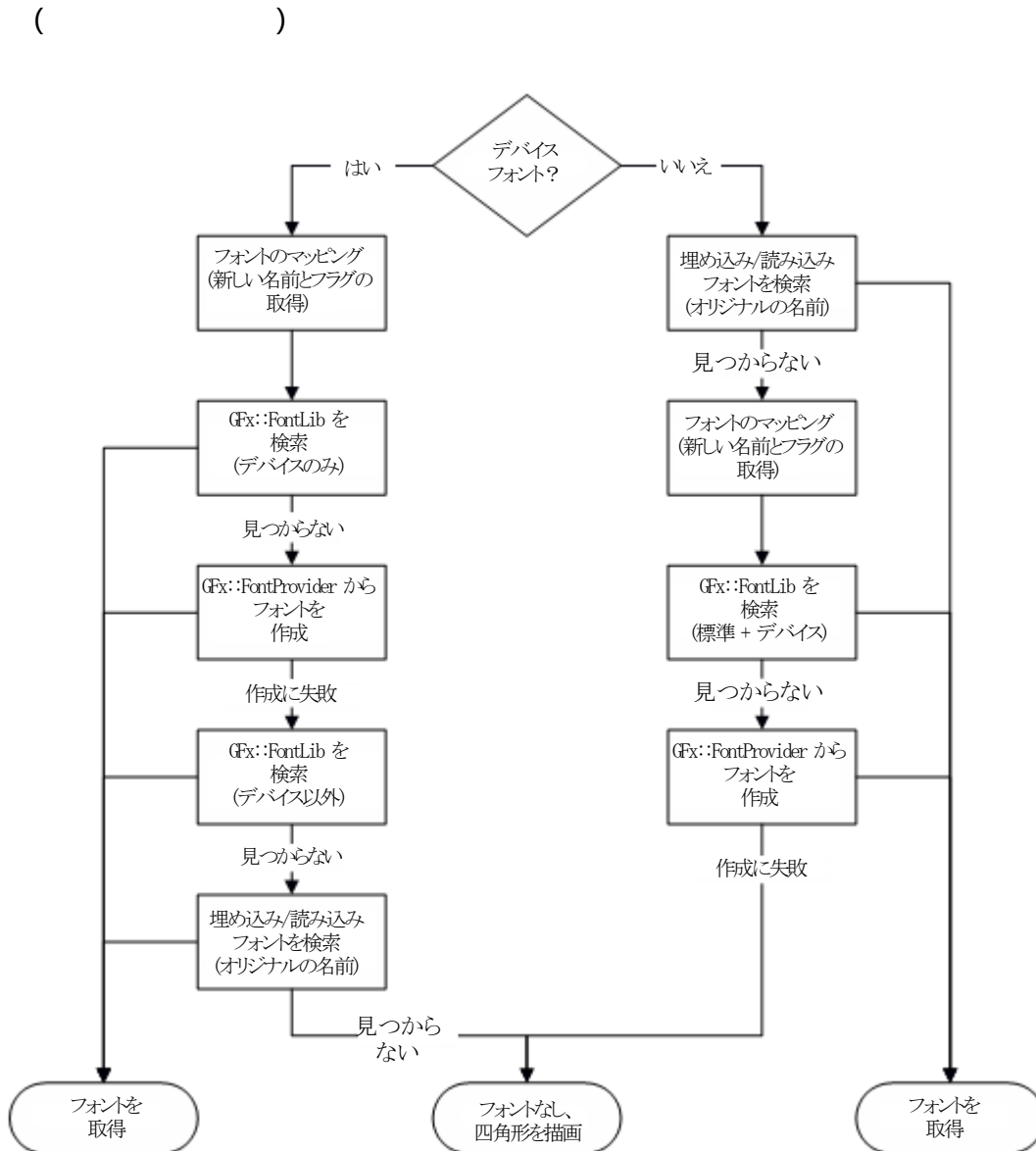
1. ローカルに埋め込まれたフォント
2. SWF/GFX
3. GfX::FontLib SWF/GFX
4. (GfX::FontProviderWin32)

Flash



4.1 フォントの参照順

Scaleform []



Gfx::FontLib Gfx::FontProvider

4.2 Gfx::FontMap

Gfx::FontMap

3

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
```

```
("TitleFont"
)
```

C++

```
#include "Gfx/GFx_FontLib.h"
. . .
Ptr<FontMap> pfontMap = *new FontMap;
Loader.SetFontMap(pfontMap);

pfontMap->MapFont("$TitleFont", "Batang", FontMap::MFF_Bold, scaleFactor = 1.0f);
pfontMap->MapFont("$NormalFont", "Batang", FontMap::MFF_Normal, scaleFactor = 1.0f);
pfontMap->MapFont("$SmallFont", "Batang", FontMap::MFF_Normal, scaleFactor = 1.0f);
```

"\$NormalFont"

"\$SmallFont"

3

MapFont

MFF_Original

scaleFactor
scaleFactor 1.0f

Gfx::FontMap

Gfx::Loader::CreateMovie

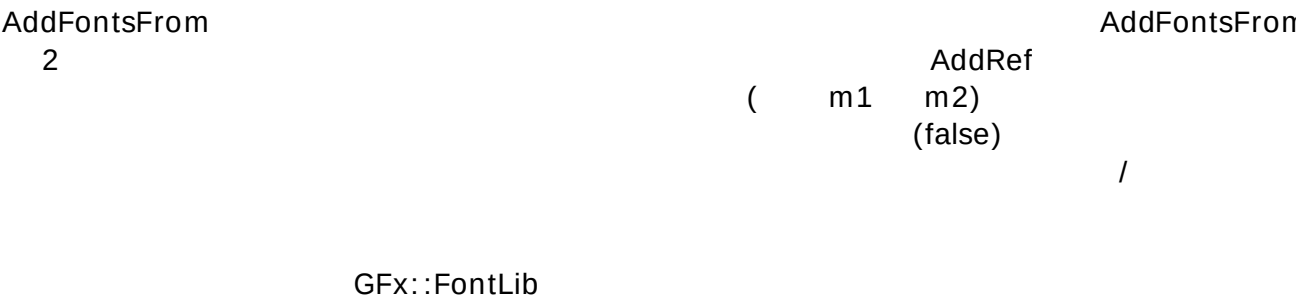
Gfx::MovieDef

4.3 GFx::FontLib



```
#include "GFx/GFx_FontLib.h"
. . .
Ptr<FontLib> fontLib = *new FontLib;
Loader.SetFontLib(fontLib);
fontLib->SetSubstitute("<default_font_lib_swf_or_gfx_file>");

Ptr<MovieDef> m1 = *Loader.CreateMovie("<swf_or_gfx_file1>");
Ptr<MovieDef> m2 = *Loader.CreateMovie("<swf_or_gfx_file2>");
. . .
fontLib->AddFontsFrom(m1, true);
fontLib->AddFontsFrom(m2, true);
```



4.4 GFx::FontProviderWin32



```
#include "GFx/GFx_FontProviderWin32.h"
. . .
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32(::GetDC(0));
Loader.SetFontProvider(fontProvider);
```



“Arial Unicode MS “

abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞
僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

Adobe FLASH

Scaleform

Win32

FreeType

```
vectorRange = Font::DontHint;  
rasterRange = Font::HintCJK;  
maxVectorHintedSize = 24;  
maxRasterHintedSize = 24;
```

4.4.2

GFx::FontProviderWin32

CJK

```
fontProvider->SetHintingAllFonts(. . .);
```

or

```
fontProvider->SetHinting(fontName, . . .);
```

```
SetHintingAllFonts ( )      SetHinting ( )      ( )
```

```
void SetHintingAllFonts(Font::NativeHintingRange vectorRange,  
                        Font::NativeHintingRange rasterRange,  
                        unsigned maxVectorHintedSize=24,  
                        unsigned maxRasterHintedSize=24);
```

```
void SetHinting(const char* Name,
               Font::NativeHintingRange vectorRange,
               Font::NativeHintingRange rasterRange,
               unsigned maxVectorHintedSize=24,
               unsigned maxRasterHintedSize=24);
```

Name UTF-8

GFx::FontProviderWin32

Windows API GFx::FontProviderWin32

```
void FontProviderWin32::SetRasterFormat(unsigned format, UByte* gamma=0);
```

format
GGO_BITMAP (by default),
GGO_GRAY2_BITMAP,
GGO_GRAY4_BITMAP,
GGO_GRAY8_BITMAP

GFx::FontProviderWin32.h <windows.h>

gamma	0...255	byte	
GGO_GRAY2_BITMAP		GGO_GRAY4_BITMAP	
GGO_GRAY8_BITMAP			
GGO_GRAY2_BITMAP	0...4	GGO_GRAY4_BITMAP	0...16
GGO_GRAY8_BITMAP	0...64	gamma	5
17 65		0	255
	0 255		
	GGO_GRAY2_BITMAP		0 63 127 191
255			

4.5 GFx::FontProviderFT2

David Turner FreeType-2

GFx::FontProviderWin32

FreeType

()

Windows MSVC

freetype<ver>.lib – DLL
freetype<ver>_D.lib – DLL

```

freetype<ver>MT.lib – ( CRT)
freetype<ver>MT_D.lib – ( CRT)

“<ver>” FreeType ( 2.1.9 219)

freetype2/include freetype2/objs

```

Gfx::FontProviderFT2

```

#include "Gfx/Gfx_FontProviderFT2.h"
. . .
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
<Map Font to Files or Memory>
Loader.SetFontProvider(fontProvider);

```

このコンストラクタは、一個の引数を持ちます。

```
FontProviderFT2(FT_Library lib=0);
```

```

FreeType ( )
FreeType FreeType Scaleform

```

```
(FT_Done_FreeType)
```

```

Gfx::Loader ( ) FreeType

```

```

fontProvider->GetFT_Library() FT_Library
FreeType
Gfx::FontProviderFT2 FreeType

```

4.5.1 FreeType

```
Win32 API FreeType ( )
```

Gfx::FontProviderFT2

```

void MapFontToFile(const char* fontName, unsigned fontFlags,
                  const char* fileName, unsigned faceIndex=0,
                  Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                  Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                  unsigned maxVectorHintedSize=24,
                  unsigned maxRasterHintedSize=24);

```

"fontName"	(	"Times New Roman")
"fileName"	(	"C:\\WINDOWS\\Fonts\\times.ttf")
"fontFlags"	"Font::FF_Bold" "Font::FF_Italic" "Font::FF_BoldItalic"	
	"Font::FF_BoldItalic"	
	"Font::FF_Bold Font::FF_Italic"	"faceIndex" FT_New_Face
	0	

Win32

FreeType

4.5.2

FreeType

```
void MapFontToMemory(const char* fontName, unsigned fontFlags,
                    const char* fontData, unsigned dataSize, unsigned faceIndex=0,
                    Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                    Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                    unsigned maxVectorHintedSize=24,
                    unsigned maxRasterHintedSize=24);
```

```
FILE* fd = fopen("C:\\WINDOWS\\Fonts\\times.ttf", "rb");
if (fd) {
    fseek(fd, 0, SEEK_END);
    unsigned size = ftell(fd);
    fseek(fd, 0, SEEK_SET);
    char* font = (char*)malloc(size);
    fread(font, size, 1, fd);
    fclose(fd);
    fontProvider->MapFontToMemory("Times New Roman", 0, font, size);
}
```

()

GFx::Loader

FreeType

Scaleform

Windows FreeType

```
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
fontProvider->MapFontToFile("Times New Roman", 0,
```

```

        "C:\\WINDOWS\\Fonts\\times.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Bold,
        "C:\\WINDOWS\\Fonts\\timesbd.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Italic,
        "C:\\WINDOWS\\Fonts\\timesi.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_BoldItalic,
        "C:\\WINDOWS\\Fonts\\timesbi.ttf");

fontProvider->MapFontToFile("Arial", 0,
        "C:\\WINDOWS\\Fonts\\arial.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Bold,
        "C:\\WINDOWS\\Fonts\\arialbd.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Italic,
        "C:\\WINDOWS\\Fonts\\ariali.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_BoldItalic,
        "C:\\WINDOWS\\Fonts\\arialbi.ttf");

fontProvider->MapFontToFile("Verdana", 0,
        "C:\\WINDOWS\\Fonts\\verdana.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Bold,
        "C:\\WINDOWS\\Fonts\\verdanab.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Italic,
        "C:\\WINDOWS\\Fonts\\verdanai.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_BoldItalic,
        "C:\\WINDOWS\\Fonts\\verdanaz.ttf");
. . .
Loader.SetFontProvider(fontProvider);

```

MapFontToFile()

5 4:

Scaleform (1 2)

- 1.
- 2. (mip-map gfxexport)
- 3. (GFx::FontPackParams)
- 4.

PC Xbox 360 PS3 Wii

GFx::FontProviderWin32 GFx::FontProviderFT2

gfxexport CPU
(PSP PS2)

Scaleform

Scaleform
Render::GlyphCacheConfig Gfx::FontPackParams
Renderer2D

NULL

GFX

		Scaleform	3.x	4.0
GfX 3.3	GfXLoader	GfXFontCacheManager		
		GfX 4.0	Renderer2D	
	GlyphCacheConfig		Scaleform	GfX 4.0
		5.2		

Gfx::FontPackParams 'gfxexport'

()

34

```

    }
}
else if (Dynamic Cache is Enabled AND
        GlyphSize < GlyphCacheParams.MaxSlotHeight)
{
    Draw the Glyph as a Texture using Dynamic Cache;
    Done = True;
}

if (Not Done)
{
    Draw the Glyph as Vector Shape;
}

```

5.2 ダイナミックフォントキャッシュの使用

```

)
( 10000

```

デフォルトではダイナミックキャッシュはイネーブルされ、そのバッファを割り当てます。

```
renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

renderer

Render::Renderer2D

HAL

。

LRU (Least Recently Used)


```

Render::GlyphCacheParams gcparams;
gcparams.TextureWidth    = 1024;
gcparams.TextureHeight   = 1024;
gcparams.MaxNumTextures  = 1;
gcparams.MaxSlotHeight   = 48;
gcparams.SlotPadding     = 2;
gcparams.TexUpdWidth     = 256;
gcparams.TexUpdHeight    = 512;

renderer->GetGlyphCacheConfig()->SetParams(gcparams);

```

TextureWidth, TextureHeight -
2

MaxNumTextures -

MaxSlotHeight -

SlotPadding -
2

TexUpdWidth, TexUpdWidth -
256x512 (128K)) 256x256 128x128

LRU
1MB 4KB 4 K

/ 256 4 K 16

4K 100 256 4K 1 MB

256 2

MaxSlotHeight+2*SlotPadding 2

2-5

) (60-80%

(UI 500 2000

1

)

Flash

(

)

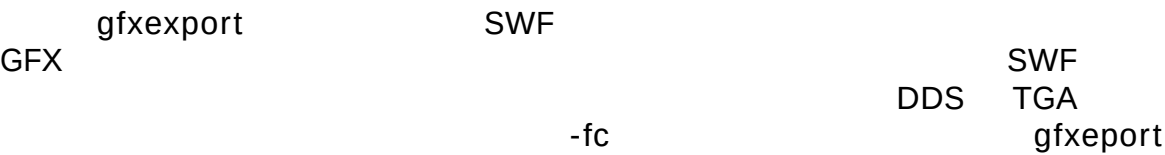
(

V

'O\W)=
 (OKOŠP@bOÁ
 Pñ=

Scaleform

5.3 フォントのコンパクター、gfxexport の使用



-fc	
-fcl <size>	Flash 256102425625%
-fcm	FontCompactor 1070% 32166412 64123216

5.4 フォント テクスチャの前処理 - gfxexport



gfxexport

CPU

Gfx::ImageCreator

Render::Renderer

mip-map ()

'test.swf' 'test.gfx'

gfxexport -fonts -strip_font_shapes test.swf

strip_font_shapes Scaleform

gfxexport

gfxexport

-fonts	()
-fns <size>	48 () 1 mip-map
-fpp <n>	3
-fts <WxH>	256x256 1 ('-fts 128' 128x128) '-fts 512x128'
-fs	

- strip_font_shapes	GFX
-fi <format>	TGA8 (grayscaled) TGA24 (grayscaled) TGA32 DDS8 (-i) TGA DDS A8 <format>

5.2

5.5 フォント グリフ パッカーの設定

GFxExport

()

Scaleform

```
Ptr<FontPackParams> packParams = *new FontPackParams();
Loader.SetFontPackParams(packParams);
```

GFxExport (.Scaleform)

GFxExport

()

```
Loader.GetFontPackParams()->SetUseSeparateTextures(Bool flag);
Loader.GetFontPackParams()->SetGlyphCountLimit(int lim);
Loader.GetFontPackParams()->SetTextureConfig(fontPackConfig);
```

SetUseSeparateTextures() "true"

"false"

SetGlyphCountLimit()

0 ()

500

(

)

SetTextureConfig()

FontPackParams::TextureConfig fontPackConfig;

fontPackConfig.NominalSize = 48;

fontPackConfig.PadPixels = 3;

fontPackConfig.TextureWidth = 1024;

fontPackConfig.TextureHeight = 1024;

NominalSize-

(

)

RAM

"NominalSize"

NominalSize

(

)

PadPixels -

1

TextureWidth、TextureHeight -

2

1

.GFX

2 Ptr<FontPackParams> packParams = *new FontPackParams();

Loader.SetFontPackParams(packParams);

...

renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));

3 Ptr<FontPackParams> packParams = *new

FontPackParams();Loader.SetFontPackParams(packParams);

Loader.GetFontCacheManager()->EnableDynamicCache(false);

()

5.6 ベクトル化の制御

Scaleform

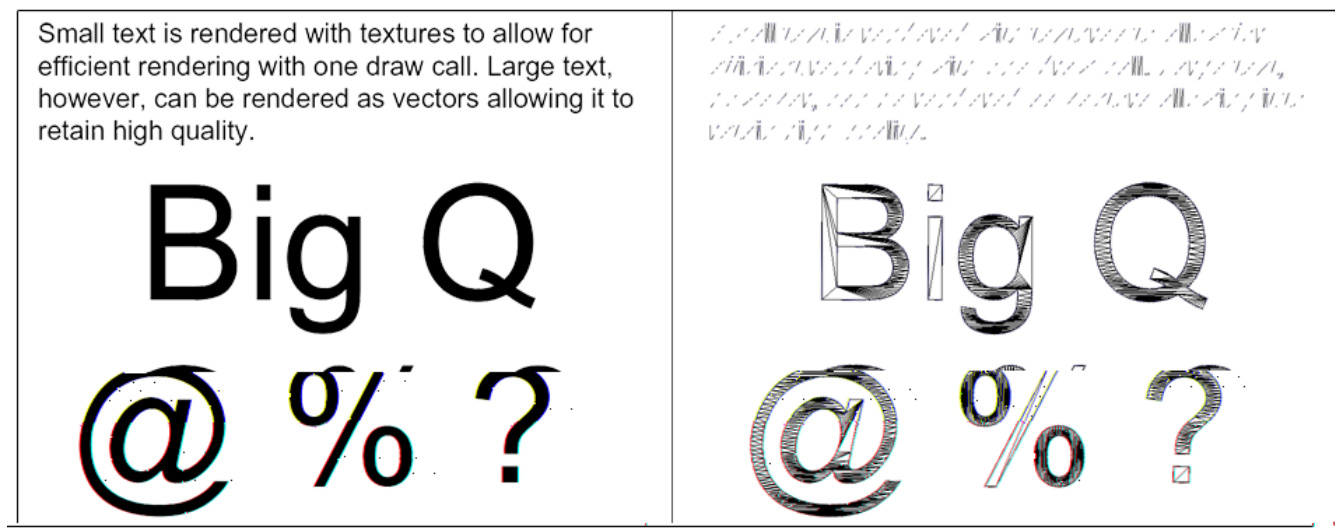
Flash

1

Scaleform

Scaleform Player

Ctrl+W



(MaxSlotHeight)
MaxSlotHeight

Render::GlyphCacheConfig::SetParams

(mip-maps)

GlyphCacheParams:: MaxRasterScale
SetMaxRasterScale

()

MaxRasterScale 1 1.25
1.25

Player
48

60

MaxRasterScale

strip_font_shapes

gfxexport

GFX

Scaleform

Player

Scaleform
Adobe Flash

Flash

Flash

Scaleform

Scaleform

6.1 フィルタの種類、使用可能なオプションと制約

Flash Studio

1



Flash

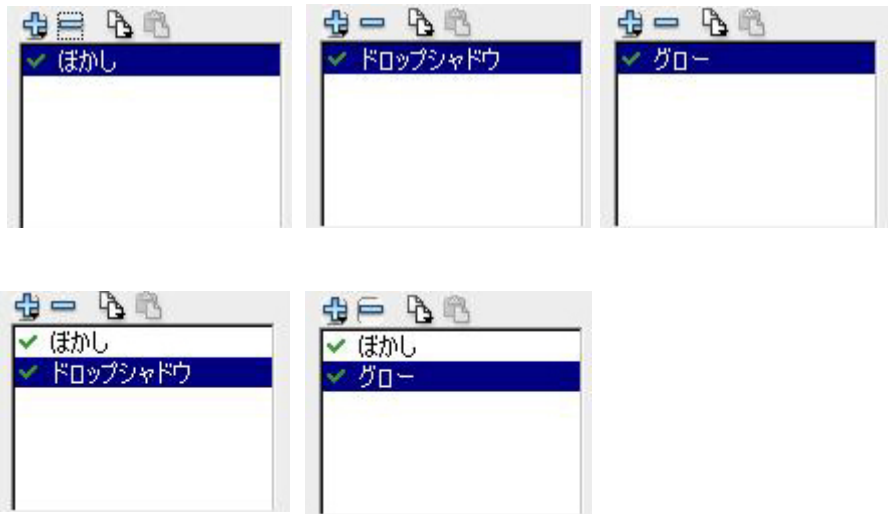
Scaleform

Scaleform
Flash

Scaleform

Flash
Scaleform

-
-
-
-



Flash

[illegible]

Blur X	Blur Y	15.9	Scaleform
			1590%

[] Flash
Scaleform

Flash GfX

Text *Text*

[])

6.2 フィルタ品質

Flash

3x3
2 []

3

[]
[] []
[] []
[]

Flash

Scaleform

Scaleform []
[:]

[]

()

Scaleform

(

)

	Low quality	Medium quality	High quality
Flash	<i>Text</i>	<i>Text</i>	<i>Text</i>
GFx	<i>Text</i>	<i>Text</i>	<i>Text</i>
	Flash	Scaleform	Scaleform

Smoothin.html <http://www.ph.tn.tudelft.nl/Courses/FIP/noframes/fip->

6.3 フィルタのアニメーション

Flash

Scaleform ActionScript

Scaleform

()
()

6.4 *ActionScript* からのフィルターの使用

Scaleform AS2 AS3 /
 DropShadowFilter BlurFilter ColorMatrixFilter BevelFilter
 Flash